

Chapter 159 — Active Surveillance of Prostate Cancer

- **A two-decade reversal:** 10–15 years ago over 90% of men in the United States, Canada and Europe with favourable-risk prostate cancer were treated radically; in some regions over 90% now start with **active surveillance (AS)**.
- **Most centres manage at least half of low-risk patients with AS**, plus growing numbers of intermediate-risk patients; variation between regions, practice types and clinicians remains large.
- **Drivers:** overtreatment concern (notably the United States Preventive Services Task Force, USPSTF); the **favourable molecular genetics of Gleason pattern 3**; long-term prospective surveillance data; better detection of aggressive disease by **imaging and biomarkers**; modelling favouring AS; above all, **patient acceptance**.

THE OVERDIAGNOSIS AND OVERTREATMENT PROBLEM

- **Sensitive testing breeds incidental diagnosis and overdiagnosis**, often through screening, in many organs.

OVERDIAGNOSIS ACROSS ORGANS IF THE ENTIRE RESERVOIR WERE DETECTED

Organ	Estimated overdiagnosis	Microfocal prevalence in healthy population
Prostate	87–94%	30–70%
Thyroid	99.7–99.9%	36–100%
Breast	43–90%	7–39%

- **Pattern:** overdiagnosis tracks the **size of the latent reservoir**, amplified by screening's bias towards **slower-growing cancers**.
- **Why prostate is especially exposed:** a **large latent reservoir** growing with age; **prostate-specific antigen (PSA)** is a cheap, simple blood test, so the **threshold to test is low**; treatment has **significant transient and permanent side effects** on quality of life (QoL).
- **PSA trade-off:** finds potentially lethal cancer at a **more curable stage** than clinical diagnosis and has **likely lowered mortality**, at the cost of diagnosing and treating **clinically insignificant cancer** before its natural history is clear.
- **USPSTF on PSA screening: 2012 Grade D** (harms outweigh benefits: overdiagnosis, overtreatment) → **2018 Grade C (neutral)**, largely because **AS uptake reduced overtreatment**.

ACTIVE SURVEILLANCE VERSUS WATCHFUL WAITING

	Active surveillance	Watchful waiting
Candidate	Fit for radical therapy if needed	Limited life expectancy (age, comorbidity)
Intent	Curative, deferred	Palliative
Monitoring	Close: serial PSA, digital rectal examination (DRE), MRI, biopsy	Limited: infrequent PSA and clinical review; no biopsy or imaging
Action	Definitive therapy on reclassification to higher risk	Palliation of symptomatic or advanced disease

	Active surveillance	Watchful waiting
Transition	To watchful waiting once life expectancy is <5–8 years (typically ~80 years, earlier with comorbidity)	—

- **Trap:** outcomes from older randomised trials and registries of simple observation reflect **watchful waiting**, not contemporary AS, in which many are reclassified and treated, probably with better results.
- **Rationale of AS:** spares radical-therapy side effects in indolent, stable or regressive disease; treats those reclassified, usually on **higher grade at later biopsy**. It rests on two premises: a **long lead time to clinical progression**, and treatment still **within the window of curability** at the first sign of higher risk.

Concept — Active surveillance defers curative intent; watchful waiting abandons it. Every difference in monitoring and trigger follows from whether the patient could still benefit from cure.

CHARACTERISTICS OF CLINICALLY INSIGNIFICANT PROSTATE CANCER

- **Autopsy prevalence of microfocal cancer ≈ age as a percentage:** 30% in the 30s, 50% in the 50s, ≥80% over 80.
- **PSA plus systematic biopsy detects cancer in 15–25%** (threshold-dependent); **lifetime risk of prostate cancer death is 3%** (North America).
- **Harbouring : death ≈ 20:1; diagnosis : death ≈ 6:1.** Most diagnosed men will not die of it and would never have been diagnosed without PSA: these "**clinically insignificant**" cancers constitute **overdiagnosis**.
- **Detection is procedure-driven:** 50–60% of men of biopsy age harbour microfocal cancer, so a **lower PSA threshold, more cores and more repeat sessions** find more insignificant cancer.
- **The challenge** is assigning an individual cancer to low risk (→ **AS**) or higher risk (→ **definitive therapy**). PSA, stage and grade help; **Gleason grade group is by far the most powerful predictor**.

GRADE TERMINOLOGY

Category	Grade group	Gleason score
Low	1	3+3 = 6
Intermediate	2 / 3	3+4 / 4+3 = 7
High	4 / 5	8 / 9–10

- **AS manages insignificant disease conservatively and reserves treatment for significant disease**, usually **grade group 1**; higher grades may still be insignificant given health and life expectancy. **Once cured by treatment, a cancer's significance can never be known.**

Molecular Genetics of Gleason Pattern 3 Versus Patterns 4 and 5

- **Gleason grading separates genetically near-normal from abnormal cells** remarkably well; the hallmarks of cancer differ profoundly between **pattern 3 and pattern 4**, explaining their different potential.
- **Six hallmarks:** sustained proliferative signalling, invasion and metastasis, angiogenesis, evasion of growth suppressors, resistance to cell death, unlimited replication. **Pattern 3 is molecularly near-normal; patterns 4 and 5 are significantly altered**, especially in key oncogenic pathways.

TABLE 159.1 GLEASON 3 VS. 4 AND HALLMARKS OF CANCER

Hallmark	Pathway or marker	Gleason 3	Gleason 4
Proliferative signalling	EGF/EGFR (epidermal growth factor/receptor)	No	Overexpressed
	AKT, MAP2 kinase	No	Aberrant

Hallmark	Pathway or marker	Gleason 3	Gleason 4
	HER2/neu	No	Amplified
Invasion and metastasis	Invasion/metastasis markers	Normal	Overexpressed
Insensitivity to growth inhibition	Cyclin D2	Expressed	Absent
Resisting apoptosis	BCL2	Negative	Strong
Angiogenesis	VEGF (vascular endothelial growth factor), microvessel density	Low	Increased
Metabolism	Reverse Warburg effect	Absent	Present
Absence of senescence	TMPRSS2-ERG	ERG normal	Increased
Tumour suppression	PTEN	Present in 90%	Deleted in 70–90%
Clinical	Metastasis/mortality after surgical grading	Absent	Present

Data from Ross et al., 2011, 2012; Skacel et al., 2001; True et al., 2006; Susaki and Nakayama, 2007; Padar et al., 2003; Fleischmann et al., 2012; West et al., 2001; Erbersdobler et al., 2010; Georgescu et al., 2016; Bismar et al., 2012; Wang et al., 2006; Trock et al., 2016; Hubbard et al., 2016.

- **What it is for:** pattern 3 lacks almost every hallmark. **Pattern:** every row normal in 3, abnormal in 4; cyclin D2 is the one abnormality of loss.
- **Trap — PTEN deletion and TMPRSS2-ERG are not exclusive to pattern 4:** rarer in pattern 3 (PTEN deleted in ~10%) but **commoner in pattern 3 cells coexisting with pattern 4** (Gleason 7): a **field defect** or **rapid dedifferentiation**. **PTEN loss alone is not metastatic; MYC amplification plus PTEN deletion** drives genomic instability and metastasis, and **both may be required**.

Natural history of grade group 1.

- **Metastatic potential effectively zero:** in large surgical series of Gleason 6, **every nodal metastasis had higher grade on re-review**; with surgically confirmed pure Gleason 6 the nodal rate was **zero**, and 20-year mortality negligible, the few deaths again showing **higher grade on re-review**.
- **But local invasion occurs:** occasional **extraprostatic extension**, **seminal vesicle invasion exceptional**; grade group 2 shows substantially more. This is **one argument for keeping the word "cancer"**.
- **Spontaneous grade progression is uncommon (~1–2% per year)**, mostly in **high-volume Gleason 6**; **10–20% progress biologically after 10 years**, so follow-up is **long term**. Progression need not mean intervention: 3+4 at 80 after surveillance from 65 may never threaten life.
- **Metastases in low-risk series = occult higher grade missed at biopsy**, present in ~30% diagnosed on 10–12-core systematic biopsy: **85% 3+4, 10% 4+3, 5% ≥8**.
- **MRI, initial vs confirmation biopsy:** MRI finds more higher-grade cancer at **initial** biopsy, but not necessarily at **confirmation**: in the **ASIST trial** (grade group 1, MRI-targeted plus systematic vs systematic alone), upgrading to grade group ≥2 was **33% vs 27%, not significant**.
- **The 2005 Gleason revision** regraded many 6s as 7s: **stage migration** improved cancer-specific survival (~26%) and helps explain metastases in older watchful-waiting series.

KEY POINTS

- Microfocal pattern 3 occurs in most men with age, in all races and regions.
- Pattern 3 has a very favourable mutational burden: oncogenic pathways normal in 3, aberrant in 4 and 5.
- Gleason score 6 (grade group 1) may invade locally but has no metastatic potential.

- Deaths in conservatively managed low-risk men reflect higher-grade cancer missed at diagnostic biopsy.

Natural History

- **Evidence spans pre-PSA and PSA eras**, so read it against its diagnostic pathway: PSA-screened cohorts carry **5–10 years' lead-time bias**; most lacked **modern AS** (close monitoring, serial biopsy); apparent low-risk progression reflects **25–30% occult higher grade**.
- **Longest follow-up, a Swedish cohort over 32 years** (mostly >70 at diagnosis, nearly all now dead): progression rate **stable for many years**; the **post-15-year rise in mortality is a small-numbers artefact**; no close monitoring.

THE THREE RANDOMISED TRIALS OF INTERVENTION VERSUS OBSERVATION

	SPCG-4 (Scandinavian Prostate Cancer Group 4)	PIVOT (Prostate Cancer Intervention versus Observation Trial)	ProtecT (Prostate Testing for Cancer and Treatment)
Setting	Scandinavia, pre-PSA	US Veterans Administration, 1994–2002	UK, all PSA-detected
Population	695, mean age 65; 5% PSA-detected, 88% palpable , median PSA 13: intermediate–high risk	731 (15% Black); median PSA 7.8, 50% T1c: more favourable	1643; >½ intermediate/high risk
Arms	Prostatectomy vs watchful waiting	Prostatectomy vs observation	Active monitoring vs prostatectomy vs radiotherapy
Follow-up	23 yr	Median 18.6 yr	15 yr
Result	Prostate cancer death 19.6% vs 31.3% , relative risk 0.55 ; +2.9 life-years	All-cause death 68% vs 73%, hazard ratio (HR) 0.84 ; absolute reduction 5.7 points (CI crosses zero); ~1 year gained	No difference in cancer mortality (all <3%) or survival ; metastases 9.4% vs 4.5% vs 5.0%
Verdict	Benefit in higher grade and age <65	Modest , mainly intermediate risk	Mortality-safe at 15 yr, more metastases

- **SPCG-4**: after prostatectomy, 3+3 and 3+4 had similar cancer mortality, **4+3 over five times higher**. Benefit was **concentrated under 65** (all-cause relative risk 0.62, cancer-specific 0.50; vs 0.86 [not significant] and 0.63 at ≥65); **absolute benefit more than doubled** from years 0–10 to 10–23; **confined to grade group ≥3, none in groups 1–2**, supporting surveillance for favourable intermediate risk.
- **PIVOT**: benefit mostly in **intermediate-risk, White, more-cores, healthier** men; low-risk observation cancer mortality **only 2.7%** (intermediate 11%, high 17%).
- **Cross-trial trap**: SPCG-4 **low-risk** and PIVOT **intermediate-risk** cancer mortality were equal (11%): **PSA stage migration, the 2005 Gleason change and lead time**, not biology.
- **ProtecT, closest to modern AS**: metastasis excess appeared by **10 years** (6.3 vs 2.4 and 3.0 per 1000 person-years) and **held at 15** (7.1 vs 3.5 and 3.7), yet **did not reach survival**, perhaps owing to **better systemic therapy**. Explanations: **23% intermediate/high grade** plus high PSAs (the >½ suffices to explain it); **77% low grade** left it **underpowered** for mortality; **active monitoring ≠ AS** (no scheduled biopsy, no defined triggers). At 15 years **61%** of the monitoring arm had been treated and **24.4%** were alive and untreated; **QoL same or better in every domain**.

Concept — Treatment benefit concentrates where grade is higher and the patient younger; the lower the risk, the smaller the benefit. That gradient is the evidential basis for surveillance.

OUTCOMES OF ACTIVE SURVEILLANCE

- **Ten prospective phase II series, ~10,000 men**, several over 15 years; the **Global Action Plan Prostate Cancer Active Surveillance (GAP3)** database pools **25 centres, >15,000 patients**.
- **Cancer mortality is rare in every cohort; cardiovascular disease is the commonest cause of death**; in Toronto (median 9 years) **non-prostate death was 10× prostate death**.

INCLUSIVE VERSUS RESTRICTIVE SELECTION

	Toronto (inclusive)	Johns Hopkins (restrictive)
Entry	All Gleason 6 + selected intermediate risk (Gleason 7 or PSA >10)	Epstein criteria: Gleason 6, ≤2 positive cores, no core >50%, PSA density <0.15
Eligible at diagnosis	~50%	~20%
15-yr cancer mortality	5% (actuarial)	0.5%
Risk signal	Intermediate vs low metastasis HR 3.75 ; baseline Gleason 7 → ≥20% metastasis at 15 yr despite radical treatment on "progression"; PSA >10 barely predictive	—
Intervention	~25% at 5 yr	~25% at 5 yr

- **Lesson:** inclusion criteria set mortality; **intervention is constant at ~25% at 5 years**.

TABLE 159.2 RESULTS OF MATURE ACTIVE SURVEILLANCE COHORTS

Cohort	n	Median FU (yr)	Freedom from treatment	bNED after deferred treatment	Cancer mortality	Other-cause mortality
Toronto (Klotz et al., 2015)	993	8.5	70% at 5 yr	47% at 5 yr	5% at 15 yr (actuarial); 1.5% (actual)	32% at 10 yr
UCSF (Maggi et al., 2020)	1450	6.4	59% at 7 yr	Not reported	<1% at 7 yr	4% at 7 yr
PRIAS, multicentre (Bokhorst et al., 2016)	5302	Not reported	27% at 10 yr	94% at 10 yr*	<1% at 10 yr	11% at 10 yr
Miami (Soloway et al., 2008)	230	2.6	85.7% at 5 yr	No recurrences	0	No data
Johns Hopkins (Tosoian et al., 2020)	1818	5	48% at 15 yr	Not reported	0.1% at 15 yr (text: 0.5%)	28% at 15 yr
Royal Marsden (Selvadurai et al., 2013)	471	5.7	70% at 5 yr	85% PSA-failure-free at 5 yr	2% at 8 yr	9% at 8 yr

*Combined end point: biochemical recurrence, local recurrence, metastasis and cancer-specific death-free survival. bNED, biochemical no evidence of disease; FU, follow-up; PRIAS, Prostate Cancer Research International: Active Surveillance; UCSF, University of California, San Francisco.

- **What it is for:** cancer mortality **≤5% at 15 years** whatever the entry rules. **Pattern:** **other-cause mortality far exceeds cancer mortality**; freedom from treatment falls steadily, **about half untreated at 7–15 years**.

- **Trap — paired figures:** Toronto **5% vs 1.5%** is actuarial vs actual; Hopkins **0.1% vs 0.5%** is the table's 2020 data cut vs the text's 2015 report. Compare like with like, and adjust for **inclusion criteria**.
- **Consensus lies between the extremes: most Gleason 6 belong on AS** (not only Epstein-eligible); offer AS **selectively and cautiously for Gleason 7**. All mature cohorts **predate MRI and biomarkers**, which are broadening indications and may lower metastasis further.

KEY POINTS: OUTCOME OF CONSERVATIVE MANAGEMENT

- Three randomised trials show little or no benefit of intervention in low risk.
- Ten prospective AS series: very low metastasis; cancer mortality 0.5–5% at 15 years by eligibility.
- ~25% reclassified and treated at 5 years, similar across cohorts.
- Pre-MRI data; MRI and biomarkers should improve outcomes by earlier detection of aggressive cancer.

Concept — *Inclusion criteria set the mortality, not the intervention rate: mortality ranges 0.5–5% by who is admitted, while a quarter reclassify at 5 years everywhere.*

Patient Selection

- **Most grade group 1 (3+3 = 6) should be managed with AS:** consensus of Cancer Care Ontario, the American Urological Association (AUA), European Association of Urology (EAU), American Society of Clinical Oncology (ASCO) and National Institute for Health and Care Excellence (NICE).

VERY LOW RISK VERSUS LOW RISK

	Very low risk (Epstein)	Low risk
Definition	Grade group 1; 1–2 positive cores; no core >50%; PSA density <0.15	Any grade group 1, PSA <10
Progression to treatment	Lower	Higher; tracks with intermediate risk
Progression to metastasis	Exceptionally low	Exceptionally low

- **The split reflects the likelihood of occult higher grade** (and possibly later progression), **not volume-dependent metastatic potential**.
- **High-volume Gleason 6** (higher PSA density, more cores, greater core involvement; nomograms quantify it) raises the risk of coexistent higher grade. It **remains an AS candidate but needs scrutiny: multiparametric MRI (mpMRI) with targeted biopsy**, repeat extended biopsy and/or a **tissue genomic assay**.

TABLE 159.3 SUMMARY OF CONTEMPORARY ACTIVE SURVEILLANCE GUIDELINES

Guideline	Low risk	Intermediate risk	Tests	Other tests	5-alpha-reductase inhibitors
Cancer Care Ontario (Morash et al., 2015)	AS preferred	Active treatment; AS if selected	PSA 3–6-monthly; DRE yearly; biopsy within 6–12 mo, then 3–5-yearly	MRI if clinical and pathological findings discordant	May have a role
ASCO (Bekelman et al., 2018)	Same	Same	Same	Others investigational	No clear role

Guideline	Low risk	Intermediate risk	Tests	Other tests	5-alpha-reductase inhibitors
AUA/ASTRO (Eastham et al., 2022)	Same	AS if selected	PSA 6–12-monthly; DRE 1–2-yearly; biopsy 1–4-yearly	MRI before initial or confirmation biopsy	—
NICE (2019)	Same	Radical treatment or AS	PSA 3–4-monthly in year 1, then 6-monthly; kinetics; repeat MRI or biopsy if concern	MRI at enrolment	—

ASTRO, American Society for Radiation Oncology.

- **What it is for:** **AS for low risk is unanimous**; divergence lies in **intermediate risk** and **MRI**. **Pattern:** older guidance reserves MRI for discordance; newer **builds it in** at enrolment or before biopsy. **Traps:** **only NICE** offers radical treatment and AS as **equal options** for intermediate risk; Cancer Care Ontario's "5-alpha-reductase inhibitors may have a role" is **not supported** by the chapter (see Pharmacologic Intervention).

Watchful Waiting

- **AS presupposes fitness for radical therapy** (usually **life expectancy ≥ 10 years**), deferred only because disease is indolent.
- **Over 5 years only very aggressive high-grade disease kills**, so in elderly or comorbid men **watchful waiting** is preferred: infrequent PSA and clinical review, **no biopsy or imaging**, treatment only for **symptomatic progression** (rarely metastasis). Often used for **elderly or comorbid men with low–intermediate risk**.
- **Switch AS $\rightarrow$ watchful waiting when life expectancy is < 5 –10 years**, aided by age–comorbidity nomograms.
- **Pitfall — neuroendocrine dedifferentiation:** progression **without PSA rise**; rare with an intact gonadal axis, seen in elderly men with **unrecognised hypogonadism**. Treat **PSA as a guide**, **measure testosterone** if hypogonadism is possible, and obtain **interval imaging**.
- **No hard rule for the switch:** some with ~ 10 years' life expectancy still want treatment for significant disease; "**modus mortis**" (manner of death) motivates avoiding a cancer death; survival prediction is **inexact**; **improving systemic therapy** extends life after metastasis, which may weaken the case for definitive therapy. **Preference is central**.

Concept — *Life expectancy, not grade, chooses between surveillance and watchful waiting; the switch is a judgement because both life expectancy and the salvage value of systemic therapy keep moving.*

Management of Patients on Active Surveillance

THE TWO HAZARDS OF SURVEILLANCE

	Misattribution of grade	True biological progression
Nature	Occult higher grade missed by systematic biopsy	Pattern 3 becomes 4 or 5
Frequency	Commonest: $\sim 30\%$ of biopsy Gleason 6 upgraded at prostatectomy; similar with annual biopsy over 10 years	Much less common
Usual result	Mostly grade group 2 ; beyond group 3 rare	Many 3+4 remain indolent

	Misattribution of grade	True biological progression
Remedy	MRI, targeted biopsy, clinical predictors	Long-term surveillance

- **Predictors of occult high grade ("wolf in sheep's clothing"):** PSA density, PSA, free-to-total ratio, biopsy extent (cores, core involvement), age, race; and **restricted diffusion on MRI**, a powerful predictor.
- **Proof most upgrading is benign:** despite undergrading, pathology variability and progression, **low-risk AS mortality is extremely low**; selected **grade group 2** men are increasingly surveilled.
- **These data predate MRI. Targeted ($\pm$ systematic) biopsy undergrades far less** than systematic alone, promising **earlier detection** of aggressive cancer, **reassurance** after a negative MRI or targeted biopsy and a **more accurate intervention signal**.

Concept — The main hazard of surveillance is sampling error at entry, not biology over time; the protocol's first job is to get the entry grade right.

OTHER RISK FACTORS

Race

- **Racial differences in incidence and mortality are substantial**, from **sociodemographic factors and biology** combined. Results for Black men on AS are **inconsistent**:

BLACK RACE AND ACTIVE SURVEILLANCE

Setting	Finding
Single-institution cohort	More intervention and progression
Meta-analysis, 11 studies of AS-eligible men at prostatectomy	8: adverse pathology (margins, upgrading, upstaging); 3: no association
GAP3 consortium	Upgrading and progression worse; definitive therapy rate the same
Canary Prostate Active Surveillance Study (PASS), protocol-driven	Adverse pathology at deferred surgery similar ; race not linked to reclassification
Veterans Affairs (VA), 2004–2013	Less AS use ; more progression to definitive therapy
VA	More progression and treatment
VA, intermediate risk	No association with treatment, progression, metastasis, cancer or all-cause mortality

- **What it is for:** genuine **inconsistency**. **Pattern:** differences shrink under a **protocol-driven** strategy (PASS) versus GAP3's **heterogeneous** criteria and triggers. **Trap:** **equal access (VA) did not reliably erase disparity**; methods and outcome definitions differ, and **biology and sociodemography** are hard to separate.
- **Black men are unequivocally AS candidates:** consider **more intense surveillance**, ensure **social factors do not impede compliance**; **MRI and genomics** should bring safety comparable to White men.

Concept — Race shifts the probability of occult higher grade, not eligibility: the answer is closer scrutiny and better detection, not exclusion.

Family History

- **Family history should rarely drive radical treatment** because of "**family history inflation**": PSA-era diagnoses of insignificant cancer give relatives a "family history", so they are screened more and diagnosed with insignificant disease. **Relatives with low-grade disease remain AS candidates.**
- **Exception: metastatic disease <60 years in a first-degree relative** suggests germline predisposition → **germline testing of family, especially DNA repair genes.**
- **BRCA1, BRCA2, ATM** accelerate mutation acquisition and cause an **aggressive phenotype**; **BRCA2** localised cancers are genomically unstable, **resembling metastatic disease**. On AS, **BRCA2 carries the highest upgrading risk** (especially to grade group ≥ 3), **most upgraded within 10 years**; **BRCA1 and ATM** less so.
- **Contrast:** a small series of mismatch repair or BRCA carriers, largely diagnosed by MRI/ultrasound fusion, showed **little upgrading at 12-month confirmation biopsy.**
- **BRCA carriers: consider definitive therapy even for low grade**; initial AS is reasonable only with **very close follow-up and excellent compliance**. Consider **DNA repair testing in many patients.**
- **National Comprehensive Cancer Network (NCCN) germline testing: metastatic or high risk**; localised with **strong family history of early-onset prostate cancer** or cancers suggesting **mismatch repair syndrome**; consider for intermediate risk with **intraductal or cribriform histology.**
- **Polygenic risk:** a genome-wide association study of >5000 AS men found **18 variants predicting conversion to treatment**, and a **269-variant risk score** also predicted it; risk extends **beyond DNA repair genes.**

Concept — A family history of diagnosis is usually a screening artefact; a family history of early metastatic death, or a germline repair defect, is biology — and only biology should move a patient off surveillance.

Monitoring

- **No universal protocol; common principles: early confirmation biopsy**, ideally **high-quality MRI** with initial or confirmation biopsy, and **serial biopsy.**
- **Sequence:** MRI (if not done before diagnosis and quality MRI is available) → **confirmation biopsy** → serial PSA, DRE and MRI → **repeat biopsy while stable.**

SURVEILLANCE SCHEDULE — BODY TEXT VERSUS KEY POINTS

Element	Body text	Key points
Confirmation biopsy	Within 1–2 years	Within 1–3 years , targeting undersampled areas; omit if initial biopsy MRI-targeted
PSA	6-monthly	6-monthly
DRE	Every 1–2 years	—
MRI	No more than every 18–24 months	—
Repeat biopsy	Every 2–5 years	Every 3–5 years

- **Why they differ:** the key points incorporate a modelling study allowing **confirmation biopsy to be delayed to year 3 without harm**, and a less intensive rebiopsy interval.
- **Tailor frequency, especially of biopsy, to risk. Compliance is paramount:** compliers (especially with confirmation biopsy) had **better cancer mortality and metastasis-free survival**; published outcomes **may not apply to non-compliers.**

TRIGGERS FOR INTERVENTION

Prompts definitive therapy	Prompts evaluation, not immediate treatment
Grade group ≥ 3 , unless life expectancy significantly shortened	Increased Gleason 6 volume → MRI or molecular biomarker
Traditionally any Gleason ≥ 7 (now relaxed)	Small-volume grade group 2 , otherwise acceptable → may continue
	PSA rise alone → repeat MRI or biopsy

Concept — Grade triggers treatment; volume and PSA trigger investigation.

Magnetic Resonance Imaging

- **mpMRI (T2-weighted, diffusion-weighted, contrast-enhanced)** is sensitive and specific for higher grade. **Diffusion-weighted imaging (DWI)**: higher grade → **denser, disorganised architecture** → impeded water diffusion, so **diffusion correlates inversely with Gleason score** → MRI tends not to show low grade but defines the location, size and severity of higher grade.
- **Two uses, a pair: initial risk stratification** (strong evidence) vs **serial monitoring as sole tool** (less compelling).
- **Before confirmation biopsy**: pooled series found mpMRI **positive in most** and $\sim\frac{1}{3}$ **upgraded**, on systematic only, targeted only and both, so **both biopsy types are needed**.
- **ASIST, a pair**: **no significant difference at confirmation biopsy**, but **$\sim 50\%$ less upgrading and AS failure at 2 years**, with large **site variation** (MRI quality and reader experience matter). Another randomised trial: **MRI 3 months after diagnosis plus targeted biopsy** → **far less upgrading at 1 year**.
- **Practice**: mpMRI before confirmation biopsy, systematic plus targeted cores; recommended by NICE, an option in NCCN.
- **Serial MRI alone is insufficient**: some centres link lesion change or new lesions to upgrading or treatment, but meta-analysis shows **only moderate sensitivity and specificity** for progression to grade group ≥ 2 (negative predictive value, NPV, ~ 0.8 – 0.9): **MRI must not be the sole means of excluding progression**.
- **MRI-invisible cancers may be particularly indolent**: **visibility genes confer higher metastatic potential**, and invisible cancers have a **more favourable natural history**, supporting **"image-based" management** (target lesions as they emerge). The **value of systematic biopsy is controversial**.

Concept — Diffusion restriction tracks grade, so MRI sees what matters and misses what does not: its strength at entry, why invisibility reassures, and why it alone cannot exclude progression.

Biomarkers

- **Prospective AS cohorts exceed 10,000 men**, thousands beyond 10 years. **Clinical parameters** (PSA, PSA density, biopsy extent, race, T-stage) already stratify; clinical progression is only **0.2–5% at 15 years**, so **incremental biomarker benefit is modest**.
- **Above-average-risk patients may benefit**: **reassurance** from a favourable score, and **earlier identification** of progressors. **Access to genomic testing increases conservative management**.

THE THREE FDA-APPROVED TISSUE GENOMIC ASSAYS

	Genomic Classifier (GC; Decipher)	Genomic Prostate Score (GPS; Oncotype DX)	Cell Cycle Progression (CCP; Prolaris)
Basis	22-marker RNA expression	12 cancer genes, 4 pathways: androgen receptor, cellular organisation, stromal response, proliferation	31 cell-cycle + 15 housekeeping genes, quantitative RT-PCR

	Genomic Classifier (GC; Decipher)	Genomic Prostate Score (GPS; Oncotype DX)	Cell Cycle Progression (CCP; Prolaris)
Validated for	Metastasis after prostatectomy (HR 1.5 per 10% rise; two validation cohorts); mortality and progression after salvage radiotherapy ; on biopsy, metastasis and mortality in intermediate/high risk; adverse pathology in favourable intermediate risk	High-grade and/or non-organ-confined disease (odds ratio 1.9 per 20 points)	Death in conservatively managed men (worse if CCP >2; HR 1.7 per unit); no discrimination among low scores
AS evidence	Predicts upgrading in grade group 1, not 2 ; small model gain	Mixed : one cohort predictive; larger Canary PASS cohort no association with upgrading or adverse pathology; adds only with PSA density and PI-RADS	None specific

FDA, United States Food and Drug Administration; PI-RADS, Prostate Imaging Reporting and Data System; RT-PCR, reverse-transcription polymerase chain reaction.

- **What it is for**: none was **built for AS**; each was validated mainly in **higher-risk or treated men**. **Pattern**: AS evidence is **weakest where the tests are most requested**. **Trap**: a favourable score does not override adverse clinicopathology.

Concept — A biomarker adds most where clinical parameters leave most uncertainty; in a population already at very low risk, little uncertainty is left to resolve.

CAVEATS OF TISSUE-BASED GENOMICS

- **Genetic heterogeneity**: microdissected multifocal cancers from one gland gave **markedly different GC, GPS and CCP scores**.
- **The Bayesian problem**: a 3% prior risk of metastasis at 15 years → a **90%-specific** test labels **10%** significant → **false positives ≈ three times the true risk**.
- **Hence test only higher-risk candidates**: **high biopsy volume, high PSA density, grade group 2**.
- **Repeat testing is unhelpful**: serial GPS barely changed; **only the first test predicted upgrading**.
- **Use it to set surveillance intensity** (biopsy timing, interpretation of concerning findings, triggers) — **never as the sole basis for AS**; weigh it with clinicopathology, priorities and **ability to comply**.

KEY POINTS

- Misattribution of grade (25–30%) far exceeds true progression from pattern 3 to 4 or 5 (1–2.3% per year).
- Higher-grade risk factors: PSA density >0.15, Black race, biopsy volume (positive cores, core involvement).
- Confirmation biopsy within 1–3 years, targeting undersampled areas; may be omitted after MRI-targeted initial biopsy.
- Then PSA 6-monthly and biopsy every 3–5 years.
- MRI and targeted biopsy may replace serial systematic biopsy in very low risk; MRI NPV depends on underlying risk, so higher-risk men still need systematic biopsy.
- A visible mpMRI lesion predicts progression to treatment.

Concept — A test's value depends on the prior. In grade group 1 the prior is so low that a good assay raises more false alarms than true signals, so genomics belongs with the higher-risk candidate.

PSYCHOLOGICAL ASPECTS

- **No evidence of significant psychological harm from AS:** anxiety follows the **diagnosis, not the management**; in SPCG-4, psychological function, anxiety and depression **did not differ** between prostatectomy and watchful waiting.
- **Living with untreated cancer is still a burden**, affecting adherence: **10–15%** of large cohorts choose treatment **without a predefined indication**.
- **PRIAS:** anxiety and decisional uncertainty **lower on AS than after definitive therapy, not rising over time**; confidence grows with uneventful years. A **systematic review of >12,000 men** found **no adverse effect on well-being**. A small **exercise trial reduced cancer-specific anxiety and fear of progression**.
- **Poor copers:** chronic anxiety disorders, social isolation. **Better physical health means less distress**.

Concept — Distress follows the word "cancer", not the decision to watch it; those who struggle are predicted by prior vulnerability, not by the strategy.

LIFESTYLE MANAGEMENT

- **The urologist is often the only doctor seen regularly** — an opportunity to counsel on overall health.

Smoking Cessation

- **The single most cost-effective intervention in medicine.** Smoking may promote **aggressive disease and higher cancer mortality**; quitting appears to reduce this. The newly diagnosed patient is **uniquely receptive**; **two minutes of counselling** may change behaviour, with a longevity gain exceeding **any prostate cancer intervention**.

Exercise

- **Mechanisms:** modulated **growth factors and sex steroids**, better **myokine/adipokine** profiles and **immunity**, **androgen receptor** adaptation, **less inflammation and oxidative stress**, altered **tumour vascularisation**, **gene regulation** and **telomeres**.
- **Strenuous exertion ($\geq 6\times$ resting metabolic rate) works best.** In treated patients, **>3 hours/week vigorous activity** markedly reduced cancer mortality, **≥ 7 hours/week brisk walking** similarly; a long-term Canadian cohort agreed.
- **On AS, the ERASE randomised trial: high-intensity interval training lowered PSA and PSA velocity.** Counsel all AS patients to exercise.

Weight Control

- **Mechanisms:** altered **leptin and adiponectin** (possible tumour suppressors); **$\uparrow$ insulin-like growth factor (IGF)-1**, **$\downarrow$ IGF-binding proteins 1 and 2** (mitogenic); **$\downarrow$ sex hormone-binding globulin**, **$\uparrow$ oestrogen**; **inflammation, oxidative stress; hyperinsulinaemia**.
- **Obesity $\rightarrow$ more high-grade cancer and cancer mortality**, independent of diet, activity and clinical factors; mortality rises with each **5 kg/m² of body mass index (BMI)** and each **0.1 of waist-to-hip ratio**, perhaps **most in Black men**.
- **On AS: obesity did not affect progression at confirmation biopsy but did at later biopsies**, implicating **true progression, not initial misclassification**. In Asian men, **BMI ≥ 27.5 kg/m² predicted adverse pathology despite low-risk criteria** and more **transition zone tumours**.
- **Chain:** obesity $\rightarrow$ cancer cells take up **essential free fatty acids** $\rightarrow$ **CD8⁺ cytotoxic T cells** are starved $\rightarrow$ **impaired immunity** $\rightarrow$ faster progression.

Diet

- **Diet influences carcinogenesis, but post-diagnosis change is unproven;** data are mostly epidemiological or from treated men.
- **AS evidence conflicts: Mediterranean diet** adherence → **less grade progression** (especially non-diabetic and non-White men) vs **Canary PASS: no difference** across three diets; the **Men's Eating and Living (MEAL) randomised trial** raised vegetable intake **without changing progression at 24 months**.
- **Advise:** varied diet **rich in vegetables** (especially **cooked tomatoes, cruciferous**), **whole grains, skinless poultry, fish; limit meat, especially processed. Prostate-healthy = heart-healthy.**
- **Comprehensive lifestyle change** (vegan diet, exercise, stress management) **lowered PSA, improved treatment-free survival** and favourably altered gene expression and **telomere length**. Since **heart disease is the leading Western cause of death**, benefit may be as much cardiovascular.
- **"Holistic surveillance"** (vegan diet plus many micronutrients) is **unproven; high-dose micronutrients may harm.**

Micronutrients

- **No supplement is of proven value on AS;** most professional bodies do not recommend them. The American Cancer Society (2012): supplements are **unlikely to improve prognosis and may increase mortality.**

MICRONUTRIENTS ON ACTIVE SURVEILLANCE

Agent	Evidence	Verdict
Vitamin D	Deficiency common in elderly and northern latitudes ; low levels linked to prostate cancer; slower PSA rise in summer; 4000 IU/day D3 for a year possibly slowed progression; combined vitamin D, omega-3, curcumin, B complex lowered PSA, most with higher baseline vitamin D; phase II ProsD trial ongoing	The exception: 1000–2000 IU/day , especially with little sunlight
Capsaicin (red chilli)	Hits several cancer pathways; growth inhibition, apoptosis, cell-cycle arrest in animals	Unproven on AS
Selenium	≥140 µg/day after diagnosis → higher cancer mortality	Contraindicated
Vitamin E	SELECT → more prostate cancer than placebo	Contraindicated

- **What it is for: plausible micronutrients can harm** — the **cautionary tale. Pattern:** only vitamin D has a favourable signal. **Trap:** "natural" is not "safe".

Concept — *What protects the heart plausibly protects the prostate at no oncological cost; isolated high-dose micronutrients have repeatedly proved harmful when properly tested.*

PHARMACOLOGIC INTERVENTION

DRUGS STUDIED FOR SECONDARY PREVENTION ON ACTIVE SURVEILLANCE

Agent	Mechanism	Evidence	Adverse effects	Verdict
5-alpha-reductase inhibitors (dutasteride)	Lower androgenic load; 5-alpha-reductase deficiency → no prostate cancer	REDEEM (Reduction by Dutasteride of Clinical Progression Events in Expectant Management): 38% less "progression", entirely less grade group 1 volume progression (cytoreduction); no difference in upgrading	↓ libido, erectile dysfunction	Not to prevent "progression" or lower PSA; if a benign prostatic hyperplasia/ lower urinary tract symptom indication exists, low-grade cancer should not alter the decision
Statins	HMG-CoA (3-hydroxy-3-methylglutaryl-coenzyme A) reductase inhibition; fewer lipid rafts → less AR, EGFR, luteinising hormone receptor signalling, ↓ Akt, apoptosis; anti-inflammatory, antiangiogenic; may inhibit androgen synthesis	Mostly epidemiological/ treated cohorts: ↓ cancer mortality, ↓ PSA recurrence post-diagnosis; some negative	—	Reasonable in many for cardiovascular benefit; consider low dose
Metformin	↓ hepatic glucose output, ↑ insulin sensitivity; AMP-activated protein kinase -dependent and -independent; ↓ insulin/IGF signalling; ↓ androgen signalling via cyclin D1	Diabetics: ↓ cancer mortality in some studies, not others; MAST (Metformin Active Surveillance Trial), placebo-controlled, results due 2025	—	Not routine
Apalutamide	AR antagonist	Phase II, 240 mg/day × 90 days : many biopsies negative immediately, not sustained at 1–2 years	Fatigue, gynaecomastia	Investigational
Enzalutamide	AR antagonist	Randomised, 160 mg/day × 1 year vs AS: less progression, more negative biopsies	Fatigue, gynaecomastia	Investigational
2-hydroxyflutamide depot	Antiandrogen	Phase IIb: ↓ PSA, smaller MRI lesions , some biopsy effect	—	Investigational

AR, androgen receptor.

- **What it is for: no drug is yet indicated** for secondary prevention; statins are justified by the heart. **Trap:** "progression" included **grade group 1 volume increase**, which 5-alpha-reductase inhibitors and AR-targeted drugs reduce by **cytoreduction** — **expected, not clinical benefit**; only **upgrading** counts.

- **Why 5-alpha-reductase inhibitors are avoided:** libido effects and lingering uncertainty about slightly more high-grade cancer in REDUCE (Reduction by Dutasteride of Prostate Cancer Events) and the Prostate Cancer Prevention Trial (PCPT).

Concept — Any agent that shrinks prostatic epithelium will appear to "reduce progression" if progression includes volume; only grade end points separate disease modification from cytoreduction.

SUMMARY

- **Use surveillance visits to counsel** (also benefiting general and cardiovascular health):
 1. **Exercise regularly**, vigorously if possible.
 2. **Maintain a healthy weight**.
 3. **Stop smoking**.
 4. **Vegetable- and grain-based diet**; moderate red and especially processed meat; prefer **fish and skinless poultry**.
 5. **Limit saturated fat**.
 6. **Vitamin D 1000–2000 IU/day**, especially with reduced sunlight.
 7. **Consider a low-dose statin**.

GAP3

- **Global Action Plan Prostate Cancer Active Surveillance:** a global AS database **funded by the Movember Foundation**, linking >250 researchers.
- **Questions:** defining **low risk** and the diagnosis; **comorbidity and life expectancy**; **monitoring tools**; **triggers** for treatment or switch to watchful waiting; **criteria of AS efficacy**.
- **Aim:** **international consensus** (partly achieved), with comparison across **races and geographies**. **Protocol heterogeneity limits conclusions**; longer follow-up with imaging and biomarkers promises an international standard.

BARRIERS TO ACTIVE SURVEILLANCE

- **Uptake is relatively low in many constituencies**; US use has risen but **varies widely**.
- **Barriers:** **preconceptions about "cancer"**; **misinformation** from friends, family or practitioners; **provider bias**; **psychological factors**.
- **Remedy** — **shared decision-making**: (1) **inform** about the reasonable options; (2) **discuss** them against the patient's values. **Better-informed patients prefer conservative management**; **decision aids** for low risk are invaluable.
- **Four requisites of shared decision-making** (each rests on communication): **both parties involved** → **both share information** → **both express preferences** → **a decision both agree**.
- **The physician is navigator, not driver**; AS is a **journey, not a destination**, shaped by internal (**values**) and external (**pressure**) factors.
- **QoL effects**:
 - **Biopsy**: mostly **no or minimal erectile effect**; an association with serial **transperineal sector biopsy of 24–38 cores** suggests **more cores, more erectile dysfunction**; GAP3 shows a **small International Index of Erectile Function (IIEF) fall per serial biopsy**, of uncertain significance.
 - **Anxiety**: **low**, not increased by conservative management; exercise lowered it; **limitation: self-selection**.

- **Genitourinary QoL: as good as or better than after treatment** on validated instruments (EORTC QLQ-PR25, IIEF-15, FACT-P, EPIC), **comparable to men without cancer**; limited long-term data, and **late effects vs ageing** are hard to separate.
- **Self-discontinuation without criteria occurs in every cohort (7–15%)**; one systematic review attributed 20% of discontinuation to **choice or anxiety**. QoL follow-up then often stops, **obscuring true QoL impact**.

Nomenclature

RENAMING GLEASON SCORE 6

For renaming	Against renaming
Proposals: indolent lesion of epithelial origin (IDLE) ; prostatic neoplasm of low malignant potential (PNLMP)	Meets traditional pathological criteria for cancer ; invades locally
Less anxiety (patients fear a lethal disease)	Mutational landscape is a continuum ; some Gleason 6 carry significant mutations
Eases conservative management	With undergrading , men may abandon follow-up — especially worrying for Black and Native American men
Better retention on AS	Metastasis is not required to define cancer (basal cell carcinoma, gliomas)

- **Verdict:** "cancer" is **unlikely to be dropped**; the **grade group system** (low grade = **1 of 5**, not **6 of 10**) delivers many of the benefits.

Economics

- **Management is costly** (high prevalence × costly therapy). **AS costs less than treatment** despite long follow-up; overtreatment costs **money and QoL**. For low risk, **first-year definitive therapy costs substantially more** than no active treatment; a Canadian model (including other-cause mortality and recurrence) projected **large 10-year savings** versus treating all.
- **Costs follow test frequency and intervention rate: fewer biopsies, lower cost; biannual biopsy matches surgical cost by 10 years. Savings need retention:** 15-year retention **43% and 55%** (Toronto, Johns Hopkins).
- **MRI and biomarkers add cost but may be cost-effective** by improving selection and reducing treatment and rebiopsy; one model favoured **annual MRI with biopsy only for PI-RADS ≥4**. Results depend on **local, especially MRI, costs**.

Concept — *Surveillance saves money only if biopsies are few and patients stay on it; every cost lever points to imaging-triggered rather than calendar-driven biopsy.*

RESEARCH QUESTIONS

- **Focus has moved from overtreatment to undertreatment:** low-risk AS is supported by genetics, trials (PIVOT, ProtecT) and prospective series; with expanded criteria, the worry is now **higher-than-expected risk**. Open: **who qualifies, how and how often to test, with what** — and on progression, **definitive therapy for all or focal therapy for some?**
- **Molecular events of progression**, for example **circulating exosomes** carrying mRNA from aggressive into low-grade cells.
- **Germline repair defects:** localised cancers in BRCA carriers carry a mutational load **resembling castration-resistant prostate cancer**, making them **unreliable AS candidates**; offer AS **with caution** to **BRCA1, BRCA2 and ATM** carriers.

- **Finding the "wolf":** MRI reduces the **25–30% misattribution** substantially but not completely. Uncertain: **mpMRI reliability**, **need for systematic biopsy when MRI is normal** (by risk), **biomarker role**. Emerging: **PSMA-PET** (prostate-specific membrane antigen positron emission tomography), **high-resolution micro-ultrasound**.
- **Nomograms integrating MRI and biomarkers are urgently needed.** Unresolved: **restricted diffusion with a grade group 1 targeted biopsy** (aggressive, missed higher grade, or false-positive MRI?); **microfocal Gleason 6 with a mildly raised Prolaris or Oncotype DX score**; **radiomics** is in its infancy.
- **Intermediate risk: $\geq 20\%$ metastasis by 15 years** despite monitoring, yet **80% metastasis-free** — the task is identifying the indolent; **liquid biopsy** shows promise.
- **Interventions to test:** exercise, weight loss, less animal protein or carbohydrate, micronutrients (**pomegranate, capsaicin, lycopene**); promising micronutrients **often prove harmful**, whereas **statins and metformin** look promising.
- **Leaner follow-up:** can a **negative MRI replace systematic biopsy**; **2–4 targeted cores replace 12–14 systematic**; a **negative biomarker reduce biopsy**; how does **NPV vary with risk**? Can follow-up **stop** in predictably indolent disease despite **15–20 years' life expectancy**? Is **focal therapy** safe long term?
- **Can widespread AS rehabilitate PSA screening** for policy makers? This needs **modelling on recent data**.

CONCLUSION

- **AS is a safe, effective, widely adopted strategy** for low-risk disease, **preferred for most by the AUA and ASCO**, and the **antidote to screening-driven overtreatment**. The roles of **serial imaging and biomarkers** are still being defined.